Solving “DOOM” (25 pts - Bonus)

A file-format walkthrough that recovers a flag from custom Doom WAD map geometry without needing to finish the level manually.

1. The Challenge

The challenge supplied a custom MAP01 WAD and suggested launching it with Freedoom/GZDoom. The flag was said to be obvious at the end of the map.

2. Inspect the WAD Instead of Playing

A Doom WAD is a structured archive of map lumps. MAP01 contains VERTEXES and LINEDEFS that define the geometry. Parsing those structures lets us render the level from above.

3. Render the Map Geometry

After plotting the linedefs, one region of the map contains letter-shaped geometry. Zooming that region makes the hidden message readable.

4. Recover the Flag

```
deadface{LETS_PLAY_DOOM}
```

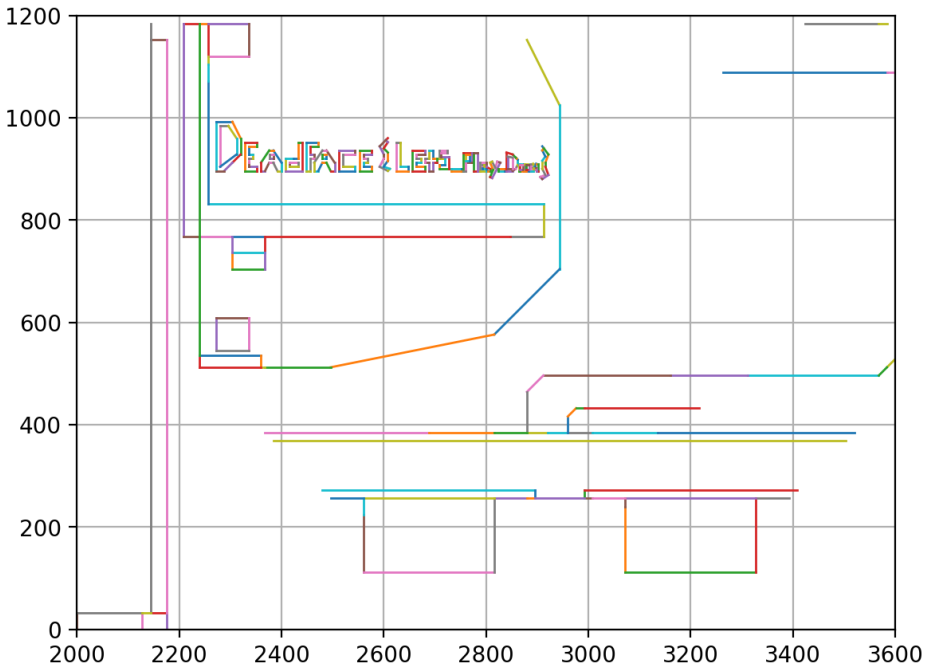
5. General Method

• List WAD lumps and locate MAP01 map data.

• Parse VERTEXES as signed 16-bit coordinate pairs.

• Parse LINEDEFS and draw each referenced vertex pair.

• Search visually unusual regions of the rendered level for text or symbols.



Rendered MAP01 region containing the letter-shaped geometry used to recover the flag.

Summary

Field	Result
Category	Bonus / file-format analysis
Format	Doom WAD
Key technique	Parse and render map geometry
Flag	deadface{LETS_PLAY_DOOM}